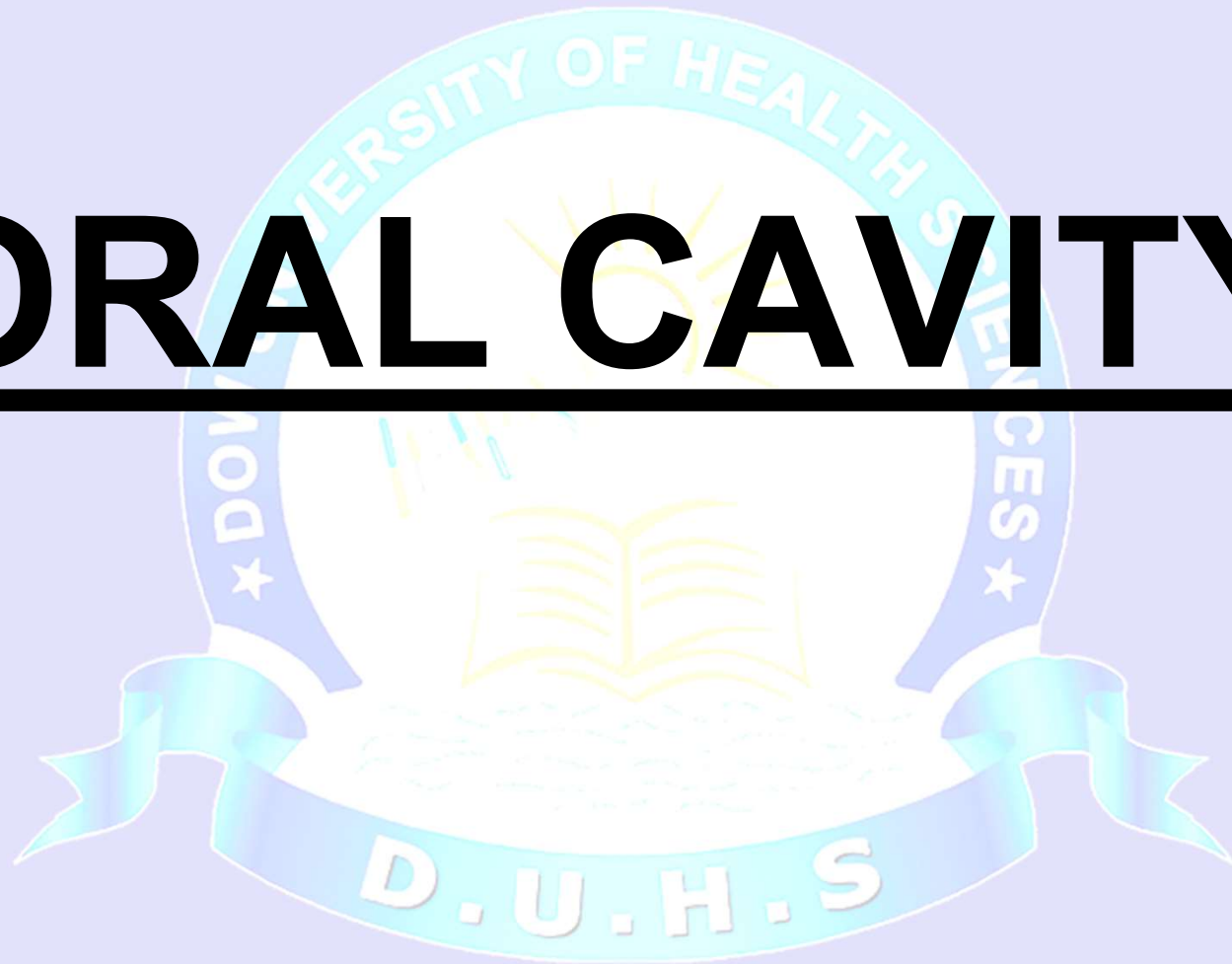




ORAL CAVITY

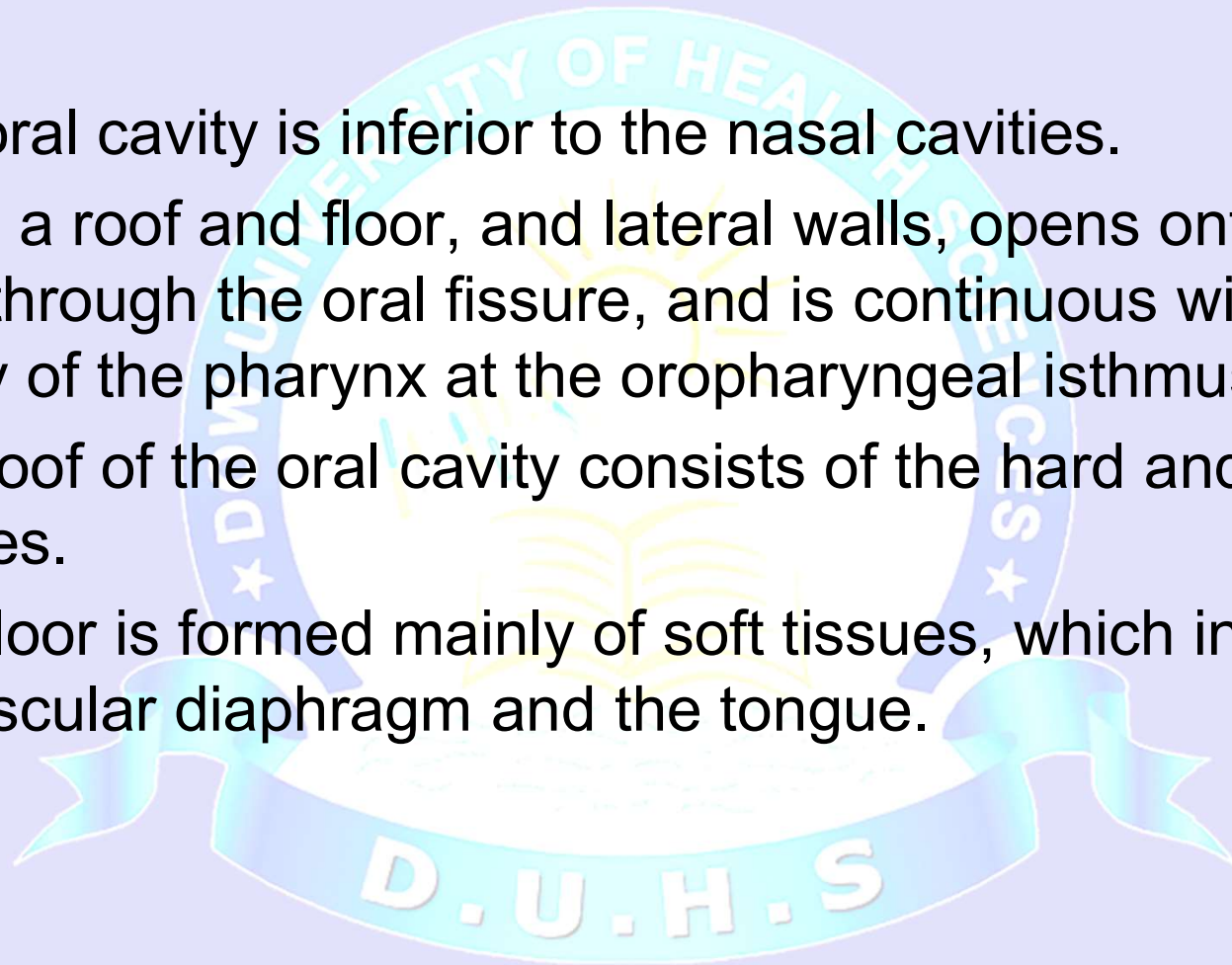


LEARNING OBJECTIVES

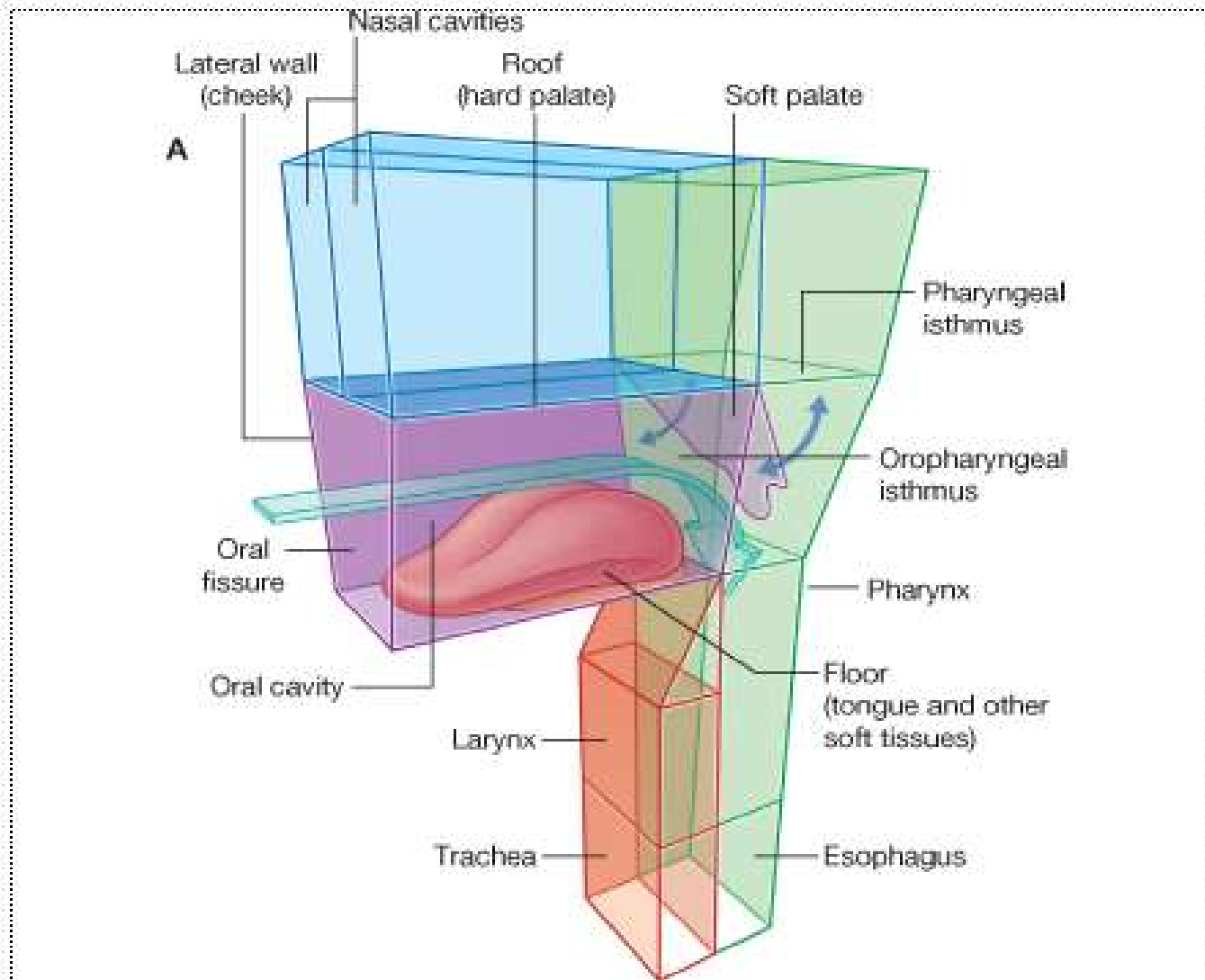
- At the end of the lecture, student should be able to know ;
- The boundaries of oral cavity
- Contents of oral cavity.
- Structures forming the roof ,lateral walls and floor of oral cavity.
- Nervous supply of oral cavity.

ORAL CAVITY

- The oral cavity is inferior to the nasal cavities.
- It has a roof and floor, and lateral walls, opens onto the face through the oral fissure, and is continuous with the cavity of the pharynx at the oropharyngeal isthmus.
- The roof of the oral cavity consists of the hard and soft palates.
- The floor is formed mainly of soft tissues, which include a muscular diaphragm and the tongue.

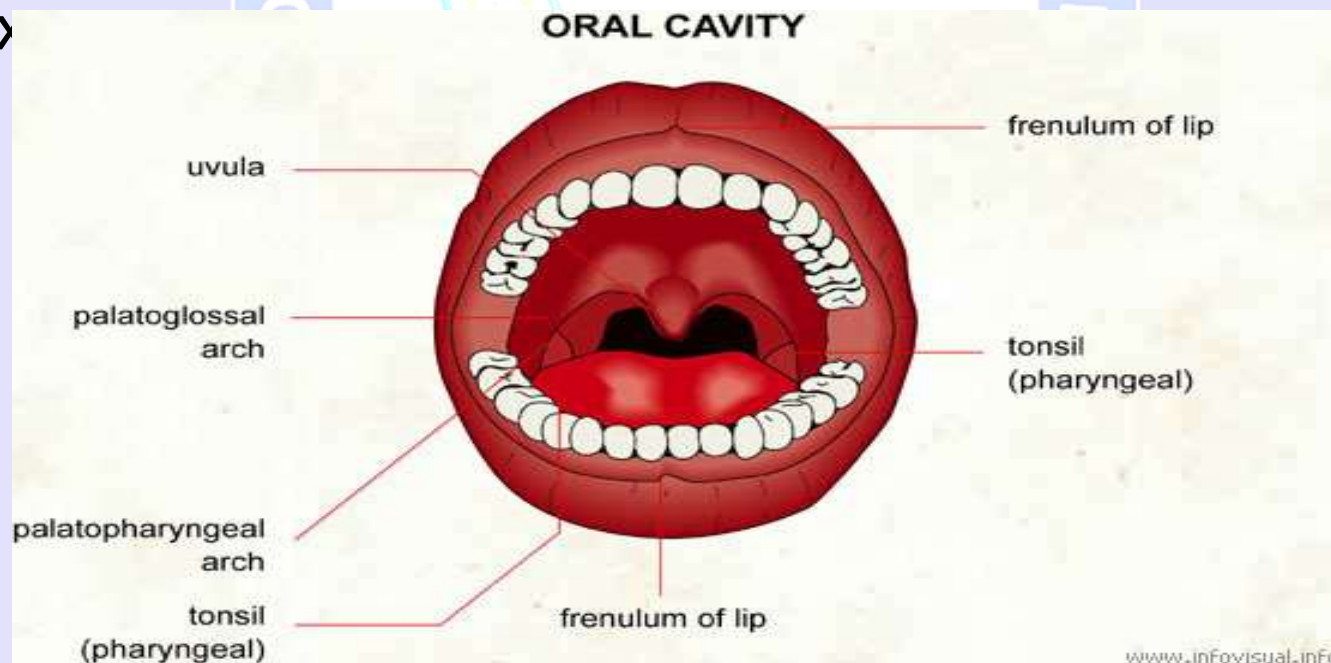


Oral cavity. Relationship to other cavities

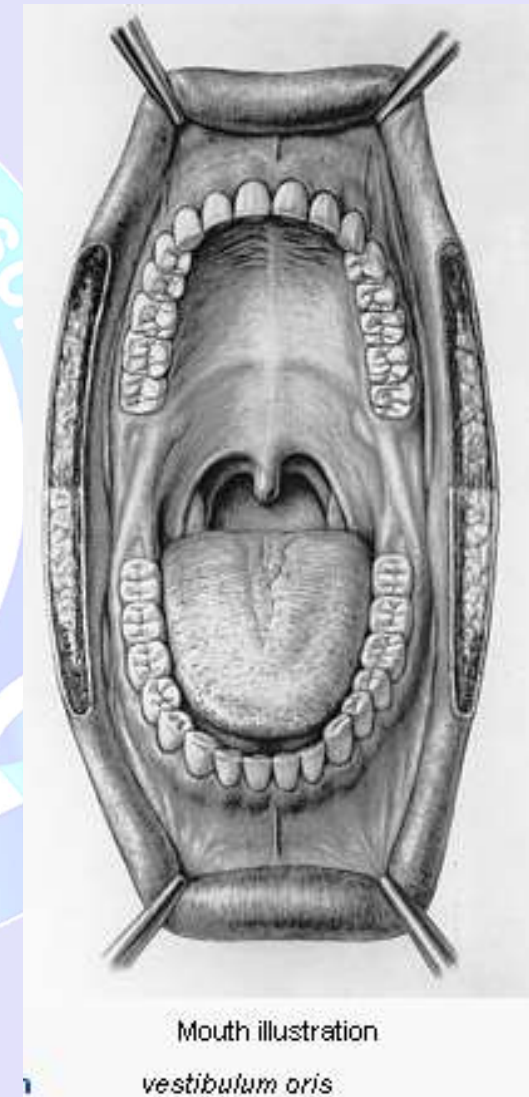
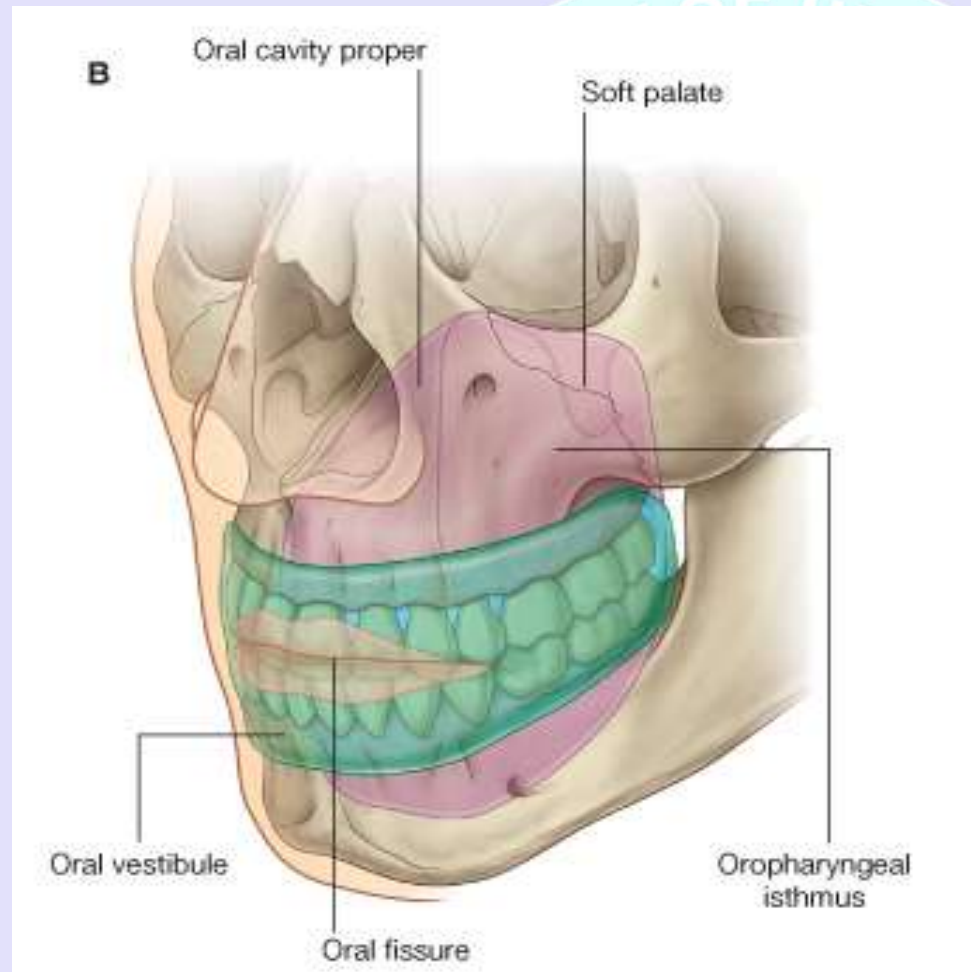


ORAL CAVITY

- The lateral walls (cheeks) are muscular and merge anteriorly with the lips surrounding the **oral fissure** (the anterior opening of the oral cavity).
- The posterior aperture of the oral cavity is the oropharyngeal isthmus, which opens into the oral part of the pharynx

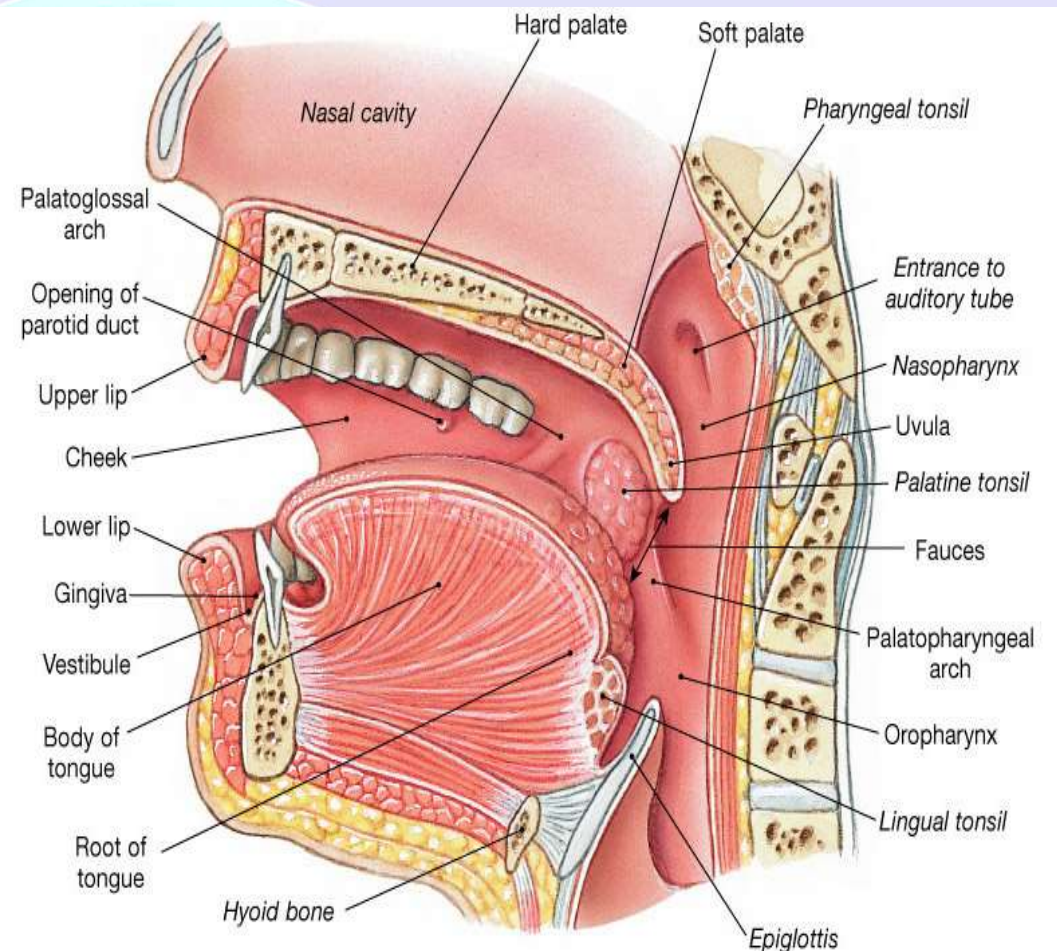


Oral cavity. Oral vestibule and oral cavity prop



BOUNDARIES OF ORAL CAVITY

- Superior
--hard and soft palates
- Inferior
--tongue and floor of mouth
- Anterolateral
--upper and lower teeth
- Posterior
—palatoglossal fold



(a) Oral cavity, sagittal section

Figure 1.

First, the boundaries of the mouth are:

- superior--hard and soft palates
- inferior--tongue and floor of mouth
- anterolateral--upper and lower teeth
- posterior--palatoglossal fold (5)

Structures to identify:

1. vestibule
2. hard palate
3. soft palate
4. uvula
5. palatoglossal arch
6. palatine tonsil
7. palatopharyngeal arch
8. posterior wall of oropharynx
9. pterygoid hamulus

Figure 2. Tongue elevated.

1. frenulum of tongue
2. ridge formed by deep lingual vein
3. sublingual fold
4. sublingual caruncle
5. opening of submandibular duct

The sublingual gland forms the sublingual fold and sends multiple small ducts into the mouth along the fold.

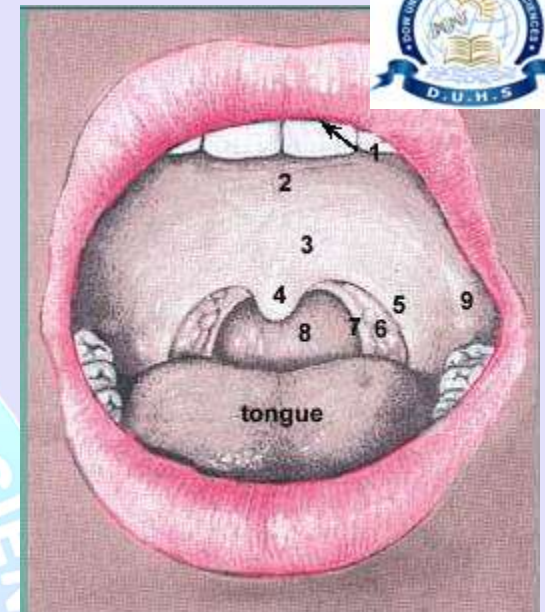


Figure 1

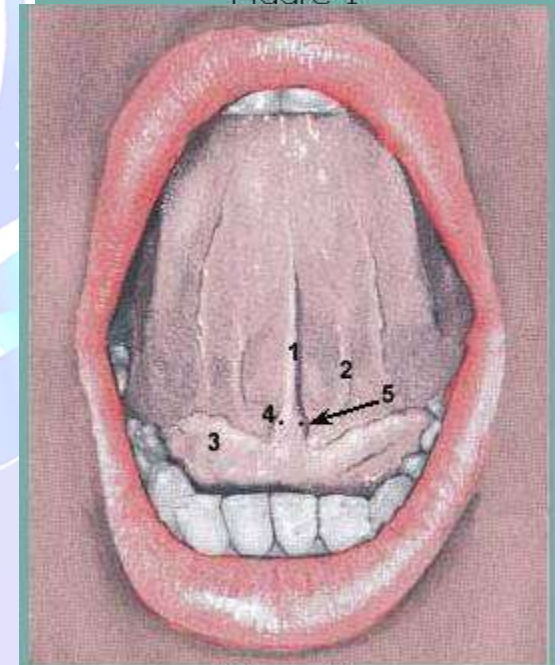
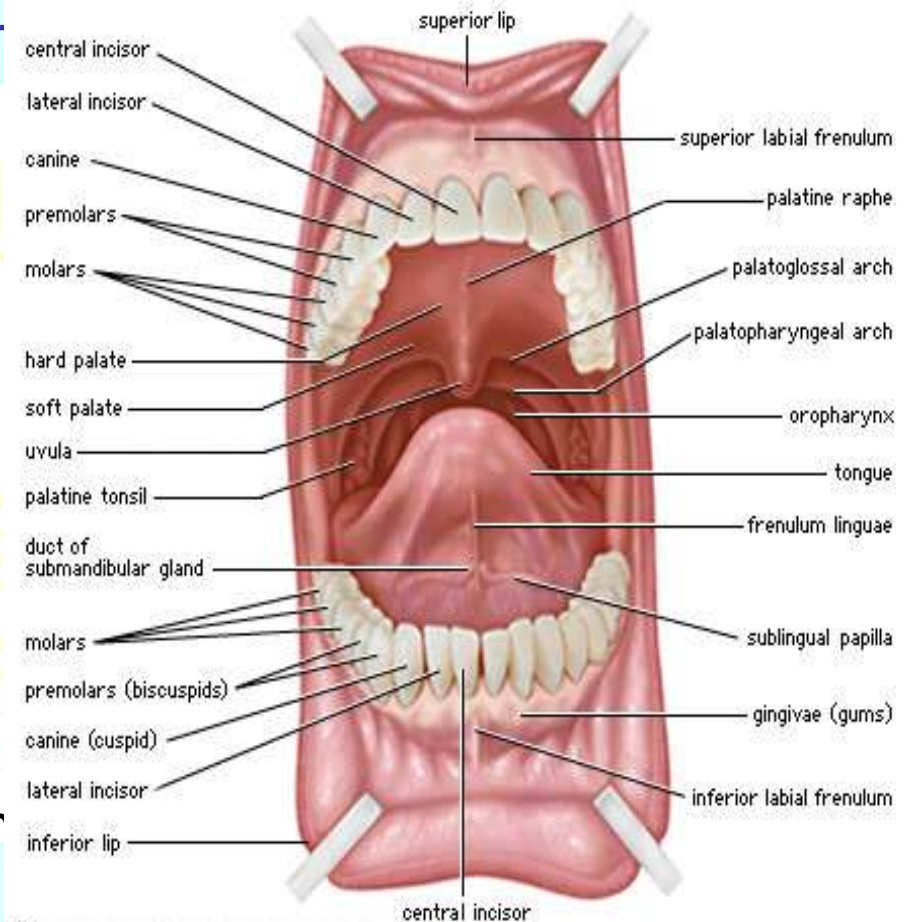


Figure 2

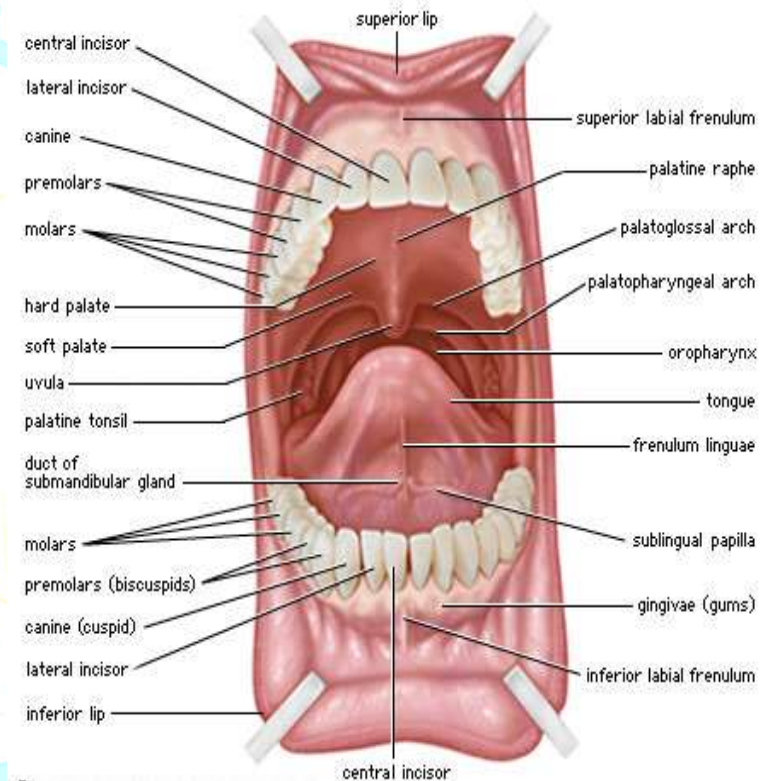
ORAL CAVITY

- ▶ The oral cavity is separated into two regions by the upper and lower dental arches consisting of the teeth and alveolar bone that supports them
- ▶ **outer oral vestibule**, which is horseshoe shaped, is between the dental arches and the deep surfaces of the cheeks and lips-the oral fissure opens into it and can be opened and closed by muscles of facial expression, and by movements of the lower jaw;
- ▶ the inner **oral cavity proper**, which is enclosed by the dental arches.



ORAL CAVITY

- The degree of separation between the upper and lower arches is established by elevating or depressing the lower jaw (mandible) at the temporomandibular joint.
- The oropharyngeal isthmus at the back of the oral cavity proper can be opened and closed by surrounding soft tissues, which include the soft palate and tongue





THE ORAL CAVITY HAS MULTIPLE FUNCTIONS

- ▶ it is the inlet for the digestive system involved with the initial processing of food, which is aided by secretions from salivary glands;
- ▶ it manipulates sounds produced by the larynx and one outcome of this is speech;
- ▶ it can be used for breathing because it opens into the pharynx, which is a common pathway for food and air.
- ▶ *For this reason, the oral cavity can be used by physicians to access the lower airway, and dentists use 'rubber dams' to prevent debris such as tooth fragments from passing through the oropharyngeal isthmus and pharynx into either the esophagus or the lower airway.*

ORAL FISSURE AND LIPS

- The oral fissure is the slit-like opening between the lips that connects the oral vestibule to the outside.
- It can be opened and closed, and altered in shape by the movements of the muscles of facial expression associated with the lips and surrounding regions, and by movements of the lower jaw (mandible).
- The **lips** are entirely composed of soft tissues.
- They are lined internally by oral mucosa and covered externally by skin.
- Externally, there is an area of transition from the thicker skin that covers the face to the thinner skin that overlies the margins of the lips and continues as oral mucosa onto the deep surfaces of the lips.
- Blood vessels are closer to the surface in areas where the skin is thin and as a consequence there is a vermillion border that covers the margins of the lips.

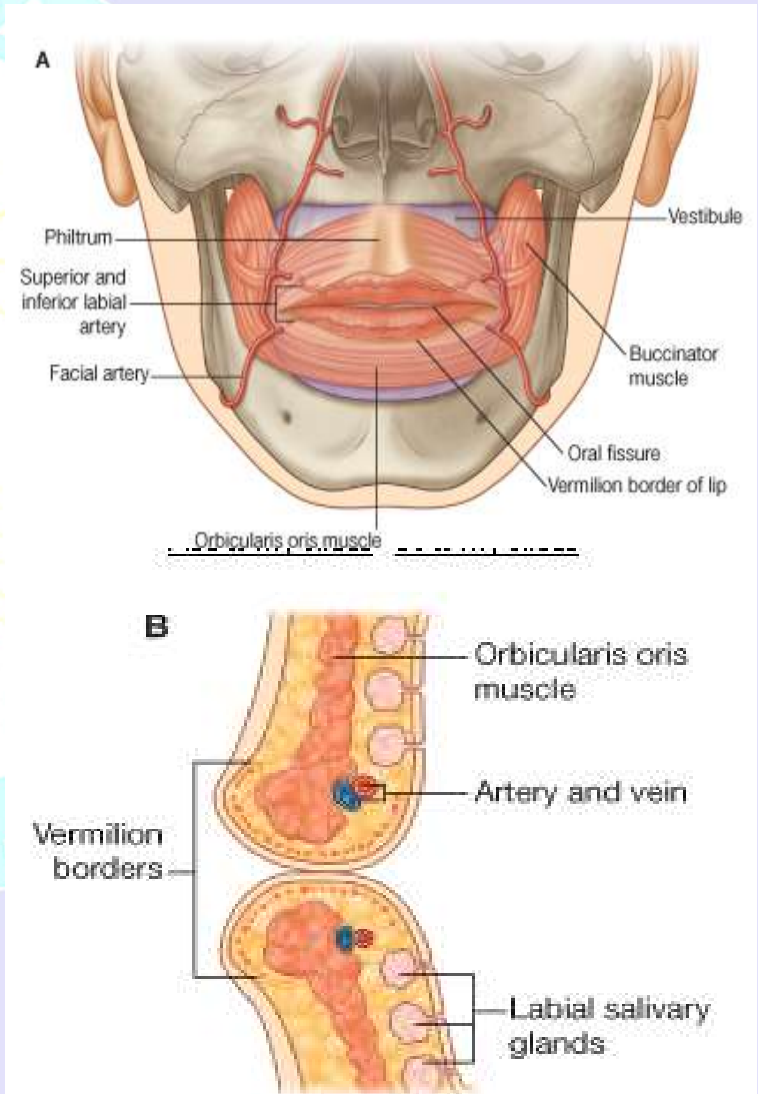
ORAL FISSURE AND LIPS

Oral fissure and lips.

A. Anterior view.

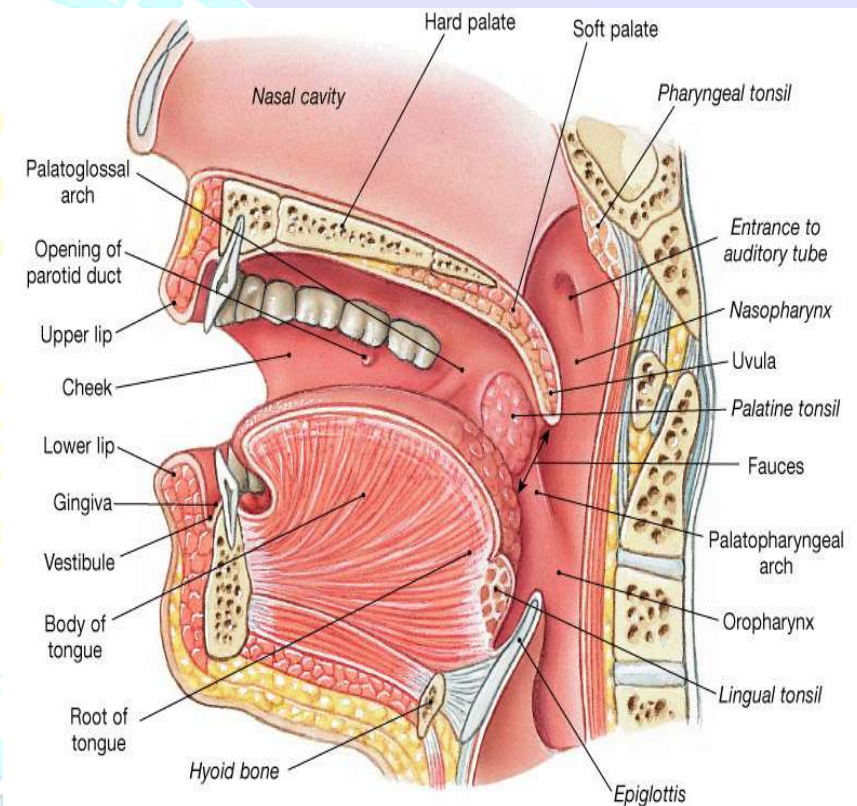
B. Lateral view.

- ▶ The upper lip has a shallow vertical groove on its external surface (the **philtrum**) sandwiched between two elevated ridges of skin.
- ▶ The philtrum and ridges are formed embryologically by fusion of the medial nasal processes.
- ▶ the inner surface of both lips, a fold of mucosa (the **median labial frenulum**) connects the lip to the adjacent gum.
- ▶ The lips enclose the orbicularis oris muscle, neurovascular tissues, and labial glands. The small pea-shaped labial glands are between the muscle tissue and the oral mucosa and open into the oral vestibule.



FLOOR OF THE ORAL CAVITY

- ▶ The floor of the oral cavity proper is formed mainly by three structures.
- ▶ a muscular diaphragm, which fills the U-shaped gap between the left and right sides of the body of the mandible and is composed of the **paired mylohyoid muscles**;

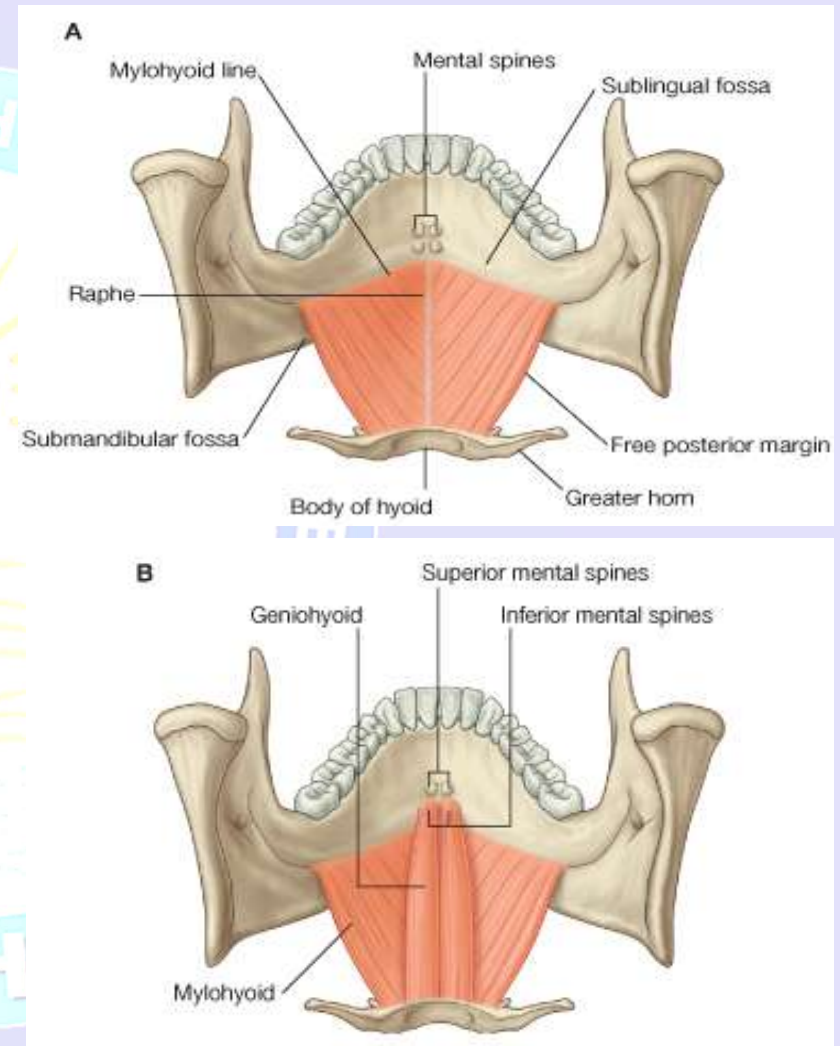


(a) Oral cavity, sagittal section

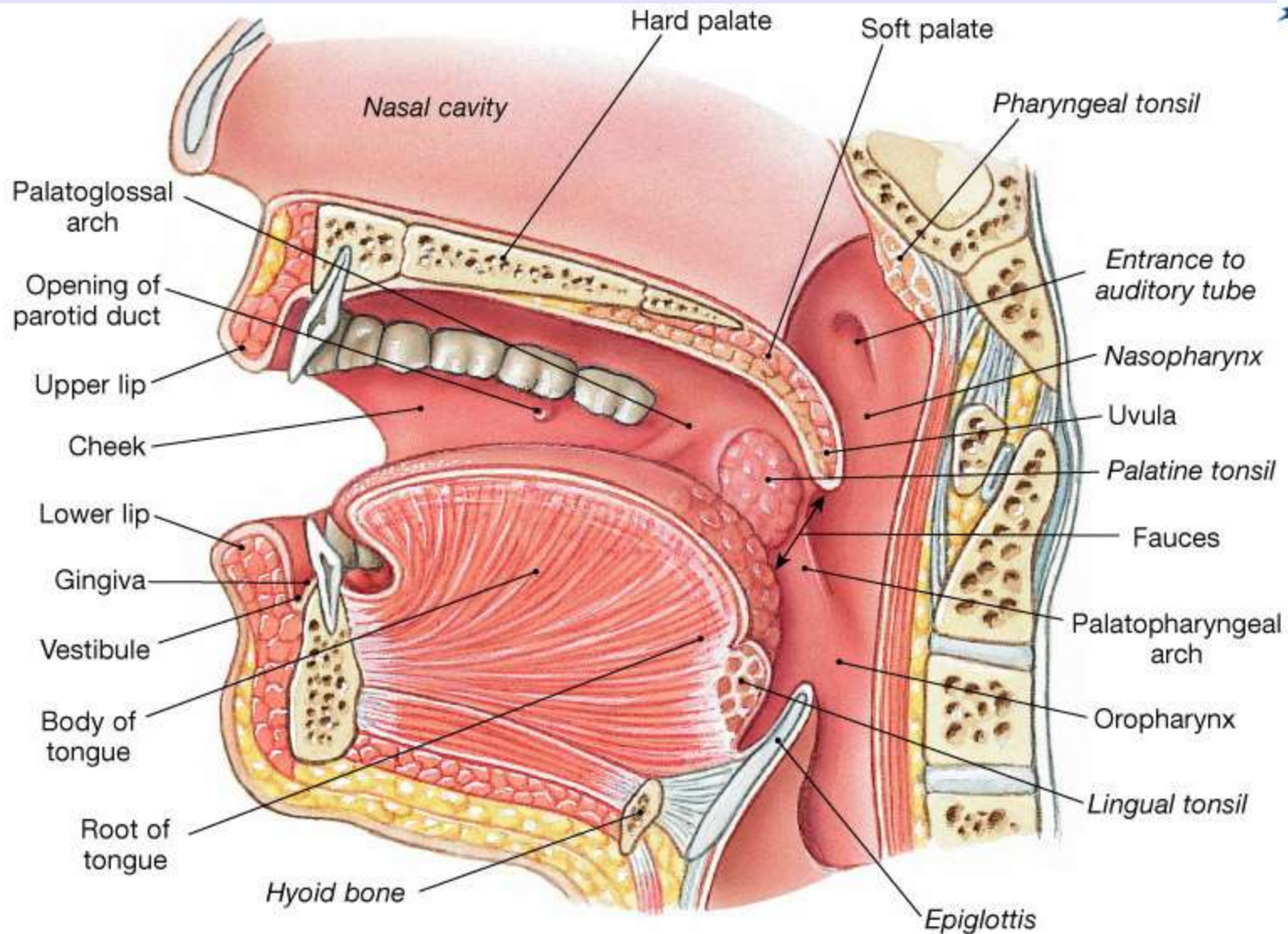
FLOOR OF THE ORAL CAVITY



- ▶ two cord-like **geniohyoid muscles** above the diaphragm, which run from the mandible in front to the hyoid bone behind;
- ▶ **the tongue**, which is superior to the geniohyoid muscles.



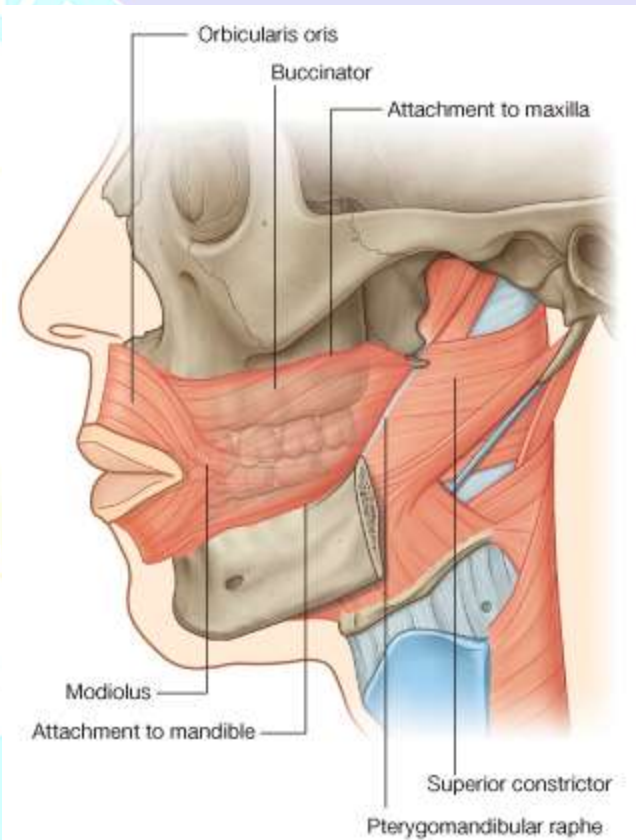
A. Mylohyoid muscles.
B. . Geniohyoid muscles.



(a) Oral cavity, sagittal section

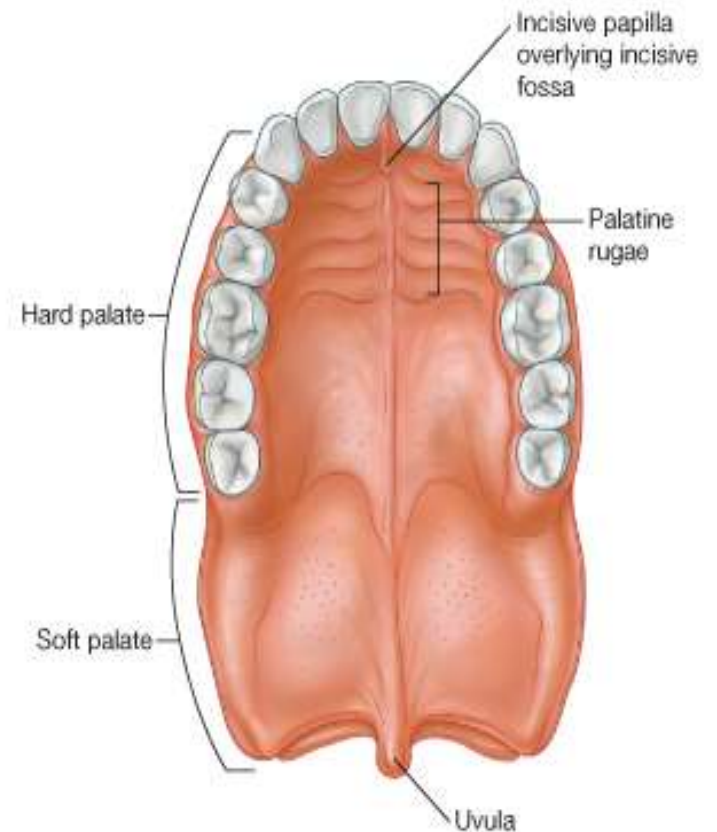
WALLS: THE CHEEKS

- ▶ The walls of the oral cavity are formed by the cheeks.
- ▶ Each cheek consists of **fascia and a layer of skeletal muscle** sandwiched between skin externally and oral mucosa internally.
- ▶ The thin layer of skeletal muscle within the cheeks is principally the **buccinator muscle**.



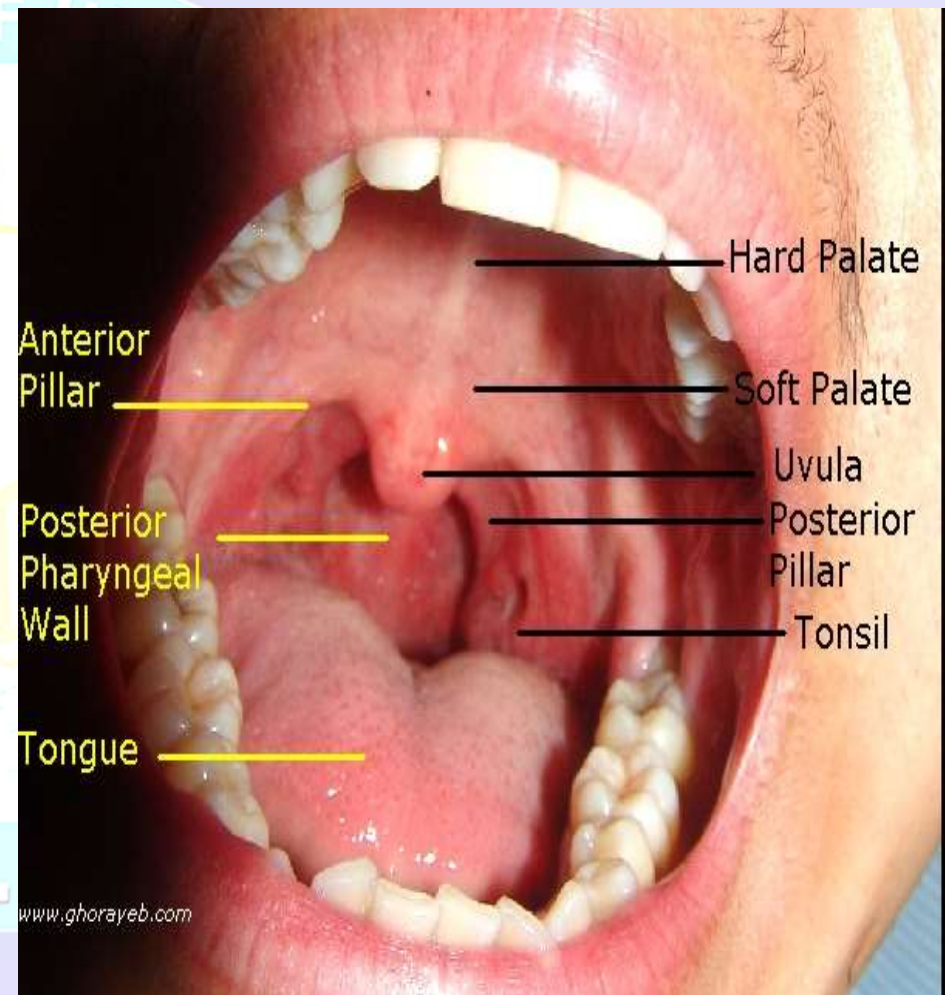
ROOF-PALATE

- The roof of the oral cavity consists of the palate, which has two parts- an **anterior hard palate** and a **posterior soft palate**.
- The hard palate separates the oral cavity from the nasal cavities. It consists of a bony plate covered above and below by mucosa: above, it is covered by respiratory mucosa and forms the floor of the nasal cavities;
- below, it is covered by a tightly bound layer of oral mucosa and forms much of the roof of the oral cavity.



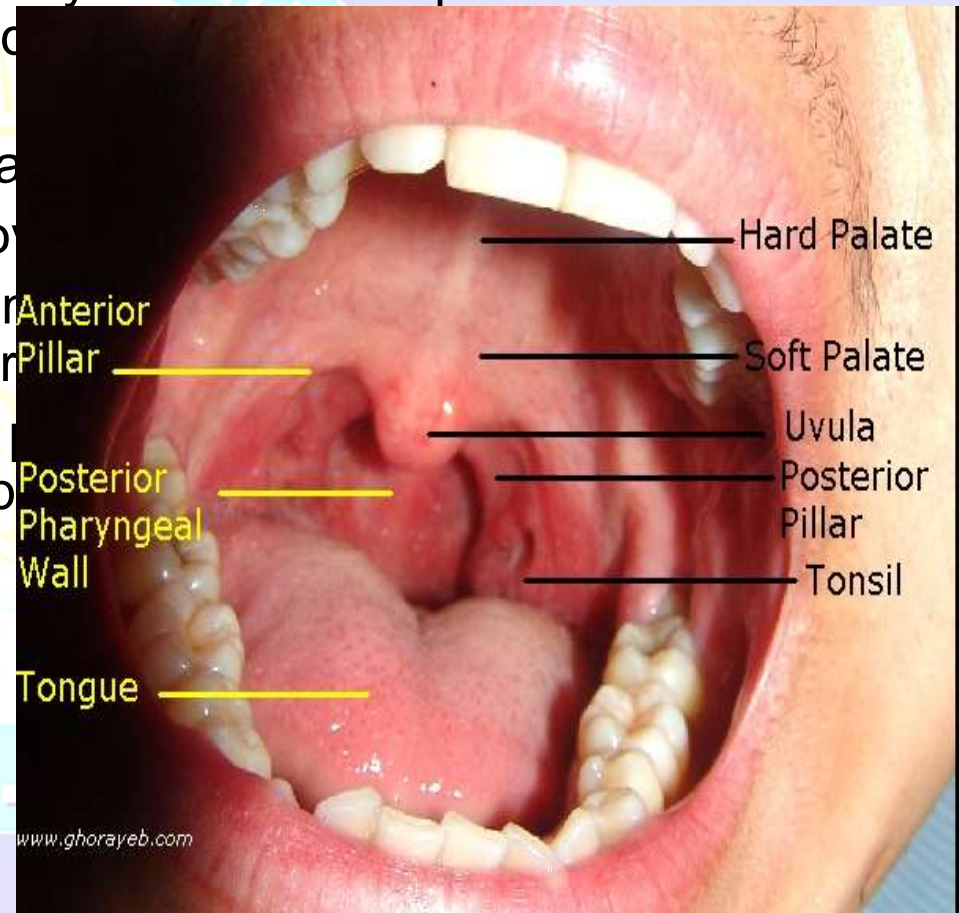
ROOF-PALATE

- ▶ The **palatine processes of the maxillae** form the anterior three-quarters of the hard palate.
- ▶ The horizontal plates of the palatine bones form the posterior one-quarter.
- ▶ In the oral cavity, the upper alveolar arch borders the hard palate anteriorly and laterally. Posteriorly, the hard palate is continuous with the soft palate.
- ▶ The mucosa of the hard palate in the oral cavity possesses numerous **transverse palatine folds (palatine rugae)** and a median longitudinal ridge (**palatine raphe**), which ends anteriorly in a small oval elevation (**incisive papilla**).
- ▶ The incisive papilla overlies the incisive fossa formed between the horizontal plates of the maxillae immediately behind the incisor teeth.



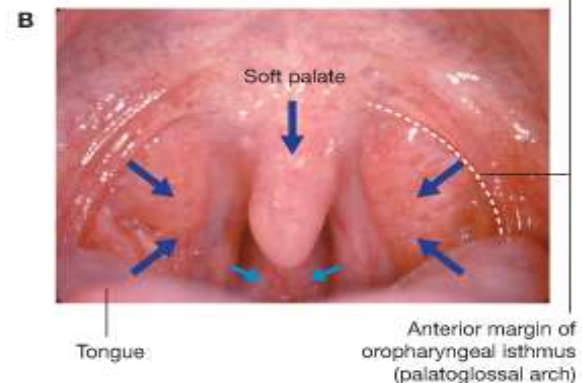
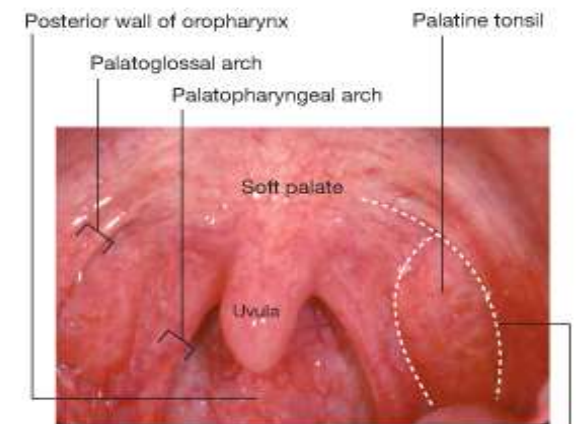
SOFT PALATE

- ▶ The soft palate continues posteriorly from the hard palate and acts as a valve that can be: depressed or elevated;
- ▶ elevated to separate the nasopharynx from the oropharynx;
- ▶ The soft palate is formed and moved by **four muscles** and is covered by mucosa lining the pharynx and oral cavity.
- ▶ The small tear-shaped muscular uvula is located at the posterior free margin of the soft palate.



PALATINE ARCHES

- ▶ The two palatopharyngeus muscles, one on each side, underlie the **palatopharyngeal arches** on the oropharyngeal wall.
- ▶ The palatopharyngeal arches lie posterior and medial to the **palatoglossal arches** when viewed anteriorly through the oral cavity.
- ▶ On each side, the palatine tonsil is between the palatopharyngeal and palatoglossal arches on the lateral oropharyngeal wall.

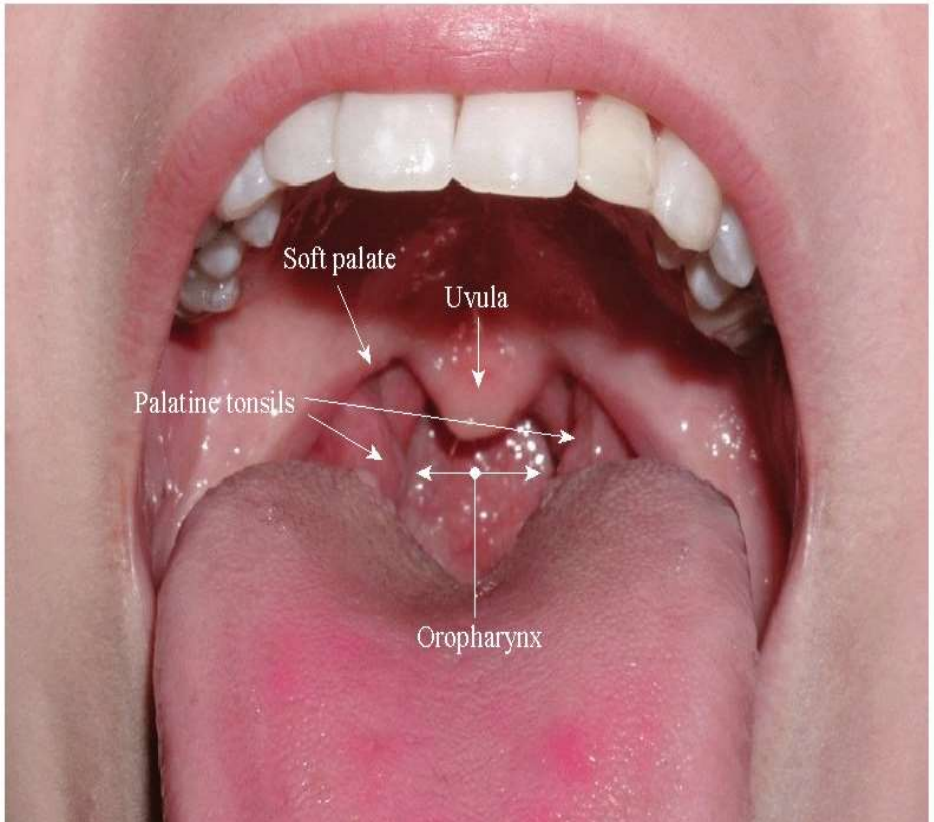


Closure of oropharyngeal isthmus

- Medial and downward movement of palatoglossal arches
- Medial and downward movement of palatopharyngeal arches
- Upward movement of tongue
- Downward and forward movement of soft palate

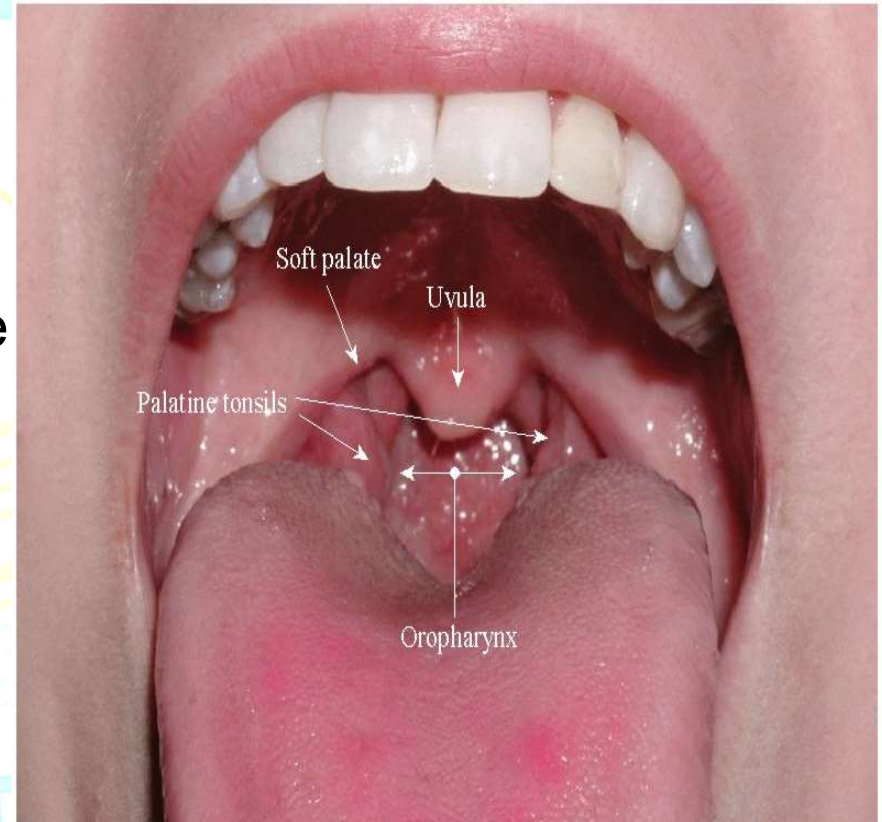
OROPHARYNGEAL ISTHMUS

- ▶ The oropharyngeal isthmus is a narrow cavity and the oropharynx. It is bounded by the palatoglossal arches;
- ▶ superiorly by the soft palate;
- ▶ and inferiorly by the sulcus terminalis which divides the oral surface of the tongue from the pharyngeal surface (



OROPHARYNGEAL ISTHMUS

- The oropharyngeal isthmus can be closed by elevation of the posterior aspect of the tongue, depression of the palate, and medial movement of the palatoglossal arches towards the midline.
- Medial movement of the palatopharyngeal arches medial and posterior to the palatoglossal arches is also involved in closing the oropharyngeal isthmus.
- By closing the oropharyngeal isthmus, food or liquid can be held in the oral cavity while breathing.

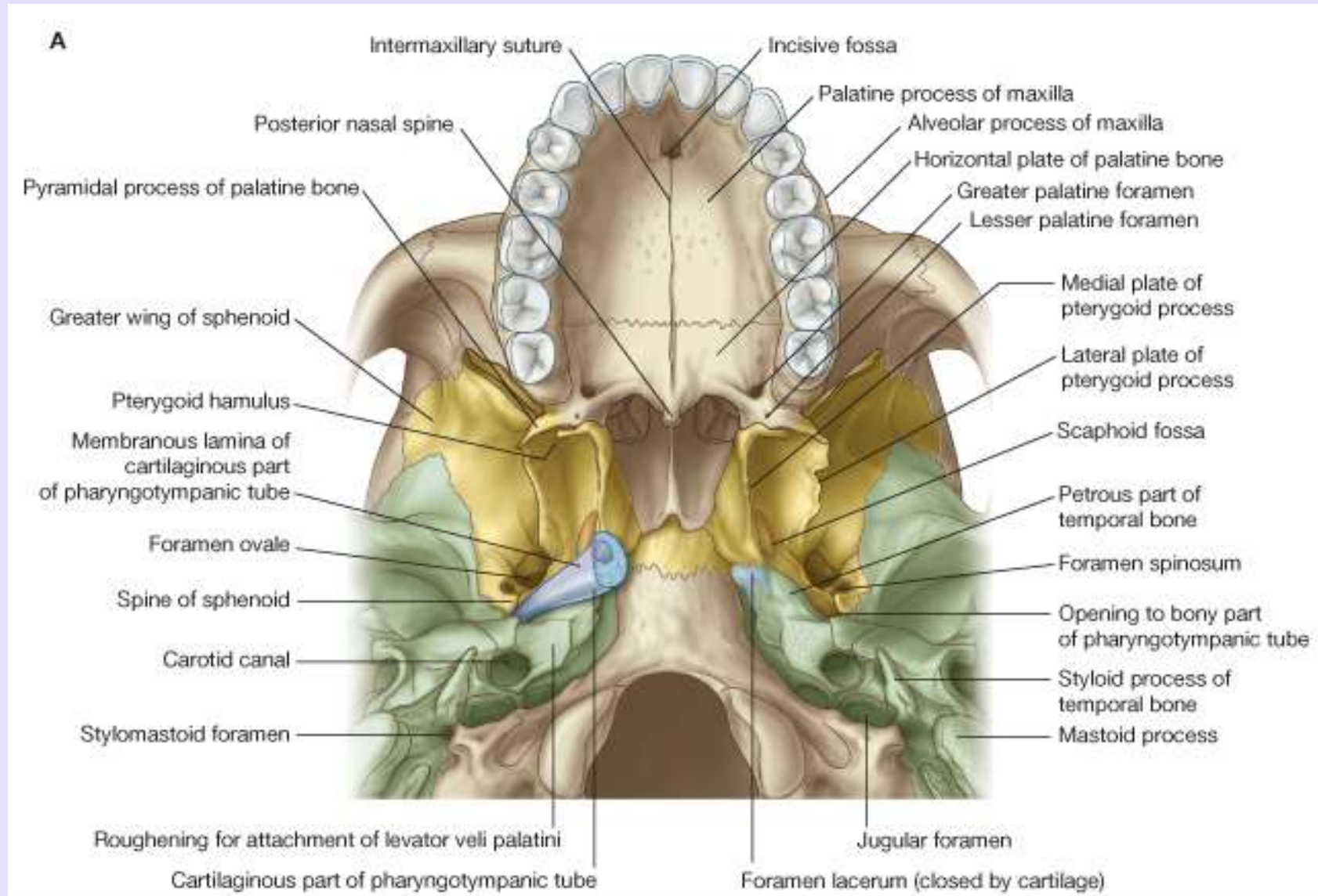




SKELETAL FRAMEWORK OF THE ORAL CAVITY

- ▶ Bones that contribute to the skeletal framework of the oral cavity or are related to the anatomy of structures in the oral cavity include:
 - ▶ **the paired**
maxillae, palatine and temporal bones;
 - ▶ **the unpaired**
mandible, sphenoid, and hyoid bone.
- ▶ In addition, **the cartilaginous parts of the pharyngotympanic tubes** on the inferior aspect of the base of the skull are related to the attachment of muscles of the soft palate

SKELETAL FRAMEWORK OF THE ORAL CAVITY



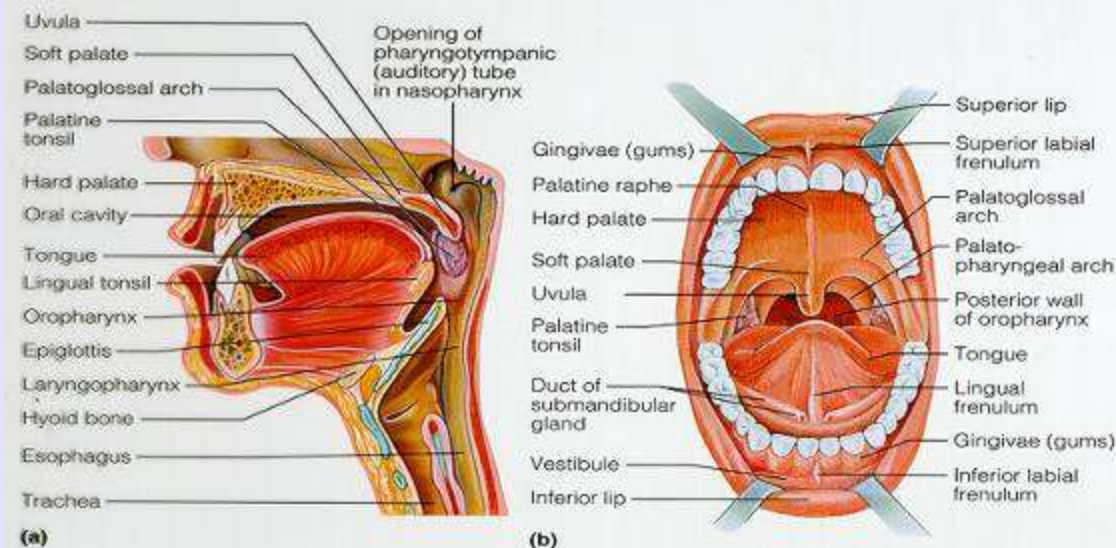
MULTIPLE NERVES INNERVATE THE ORAL CAVITY

- ▶ **General sensory innervation** is carried predominantly by branches of the trigeminal nerve [V]:
the upper parts of the cavity, including the palate and the upper teeth, are innervated by branches of the maxillary nerve [V₂];
- ▶ the lower parts, including the teeth and oral part of the tongue, are innervated by branches of the mandibular nerve [V₃];
- ▶ **parasympathetic fibers to the glands** within the oral cavity are carried by branches of the facial nerve [VII], which are distributed with branches of the trigeminal nerve [V];
- ▶ **sympathetic fibers** in the oral cavity ultimately come from spinal cord level T1, synapse in the superior cervical sympathetic ganglion, and are eventually distributed to the oral cavity along branches of the trigeminal nerve [V] or directly along blood vessels.

GATEWAY INTO THE FLOOR OF THE ORAL CAVITY

- In addition to defining the lower limit of the floor of the oral cavity, the **free posterior border of the mylohyoid muscle on each side forms one of the three margins of a large triangular aperture**, which is a major route by which structures in the upper neck and infratemporal fossa of the head pass to and from structures in the floor of the oral cavity.
- The other two muscles that complete the margins of the aperture are the **superior and middle constrictor muscles of the pharynx**.

Anatomy of the oral cavity (mouth) (Figure 24.7)



GATEWAY INTO THE FLOOR OF THE ORAL CAVITY

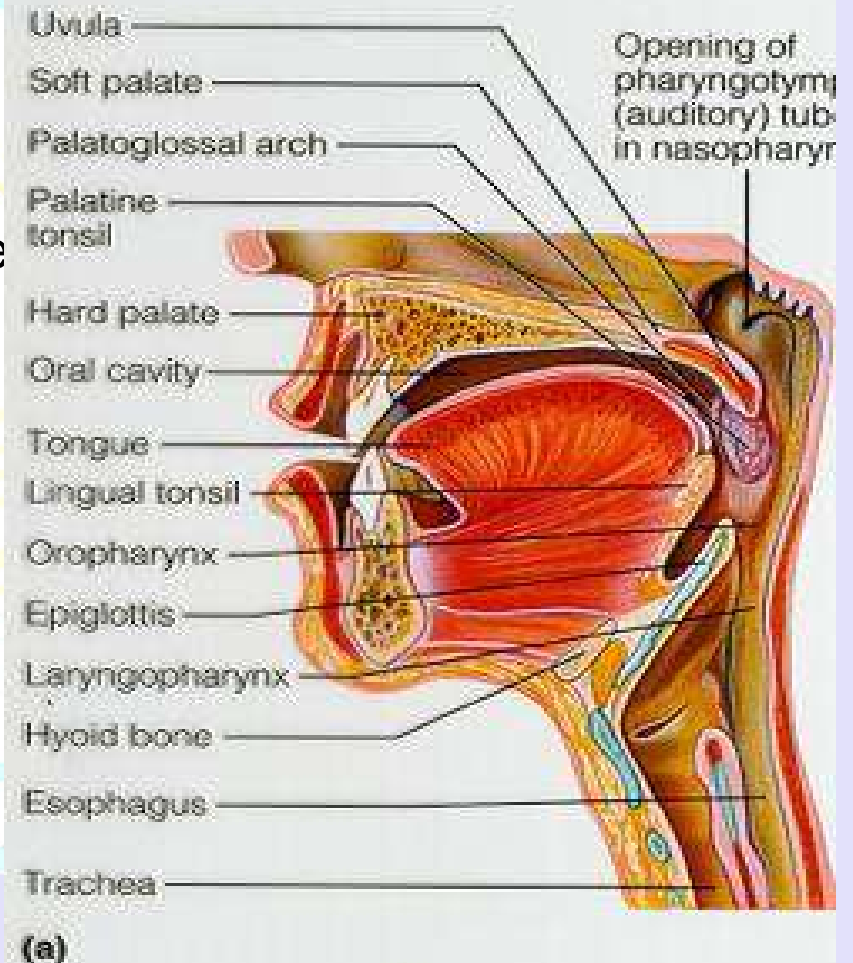


- **Most structures that pass through the aperture**

are associated with the tongue and include

- muscles (hyoglossus, styloglossus),
- vessels (lingual artery and vein),
- nerves (lingual, hypoglossal [XII], glossopharyngeal [IX]), and lymphatics.

- **A large salivary gland (the submandibular gland)** is 'hooked' around the free posterior margin of the mylohyoid muscle and therefore also passes through the opening.

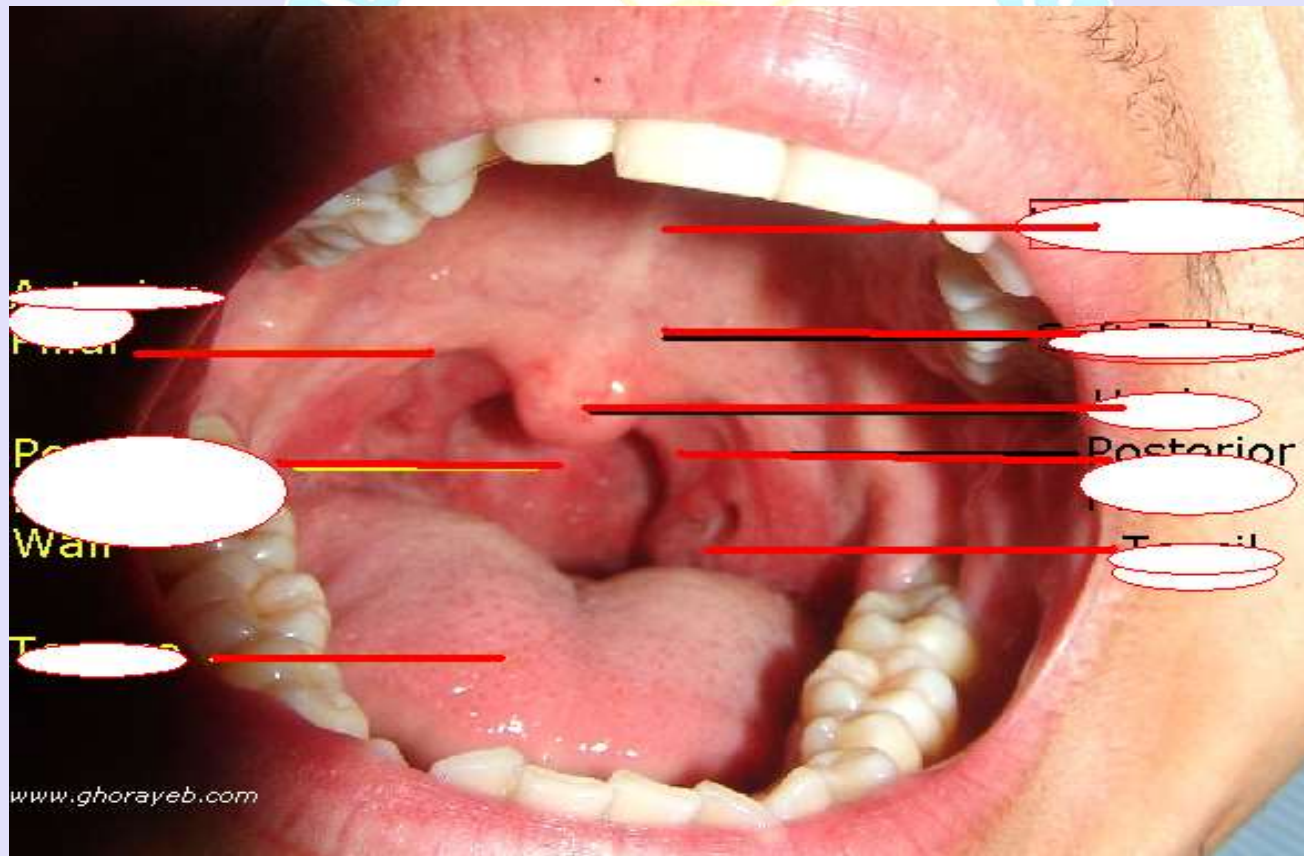


SELF ASSESSMENT

Q1.Name the structure included in oral cavity?

Q2. How can we divide the oral cavity anatomically?

Q3.Label the diagram below.



SELF ASSESSMENT

Q4.Name the structures forming the roof of oral cavity.

Q5.Name the structures forming the floor of oral cavity.

Q6.Name the nerves supplying the oral cavity (sympathetic, Parasympathetic and general Visceral)



THANK YOU

